Arista Access Node BAS-ACC-54 Basavanagudi POP Optical Transport Failure Resolution SOP (ALARM_3539)

Node: BAS-ACC-54 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3539

Description

Optical transport failures on the Arista Access Node BAS-ACC-54, situated in the Basavanagudi POP, typically manifest due to the degradation of SFP/XFP modules or fiber cuts leading to a loss of receive/transmit power. On Arista switches, this scenario is often precipitated by the inherent sensitivity of the SFP/XFP modules to environmental stressors such as temperature fluctuations, moisture, or physical shocks. Furthermore, the role of the Access Node in this context implies that the failure domain is primarily concerned with the transport of client-facing traffic to and from the aggregation network. In this specific failure domain, the ALARM_3539 is triggered when the optical signal-to-noise ratio (OSNR) drops below an acceptable threshold, causing the switch to enter an error state.

Pre-requisite Checks

- check: Verify the site location and ensure that the Basavanagudi POP is reachable., description: Confirm that the remote hands team has been dispatched to the site, if necessary.

- check: Validate the redundant link status, description: Confirm that the redundant link is up and operational before proceeding with any diagnostic or repair actions.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces tengigabitethernet 0/1/0 | include received, Expected Output: received 123456789 bytes, 0 underruns, 0 failures, 0 lost carrier, Description: This command isolates the receive statistics for the specified interface, allowing for the identification of any potential issues related to received power loss.

Command: show interfaces tengigabitethernet 0/1/0 | include transmitted, Expected Output: transmitted 123456789 bytes, 0 underruns, 0 failures, 0 lost carrier, Description: This command isolates the transmit statistics for the specified interface, allowing for the identification of any potential issues related to transmitted power loss.
```

```
Command: show log | include ALARM_3539, Expected Output: ALARM_3539: Optical signal-to-noise ratio (OSNR) has dropped below threshold, Description: This command isolates any log entries related to the ALARM_3539, providing insight into the timing and nature of the error event.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: reload replace, Description: This command initiates a reload of the specified interface, replacing the SFP/XFP module in the process. Ensure that the redundant link is up and operational before proceeding.

Step 2: Step: Verify the fiber connections, Command: show interfaces tengigabitethernet 0/1/0 | include received, Description: This command verifies the receive statistics for the specified interface, ensuring that the fiber connections are secure and functioning correctly.

Step 3: Step: Adjust the OSNR threshold, Command: configure terminal; interface tengigabitethernet 0/1/0; no osnr threshold 20, Description: This command adjusts the OSNR threshold for the specified interface, allowing for more stringent monitoring of the optical signal quality.

Post-Check Verification

```
Command: show interfaces tengigabitethernet 0/1/0 | include received, Expected Output: received 123456789 bytes, 0 underruns, 0 failures, 0 lost carrier, Description: This command verifies that the receive statistics for the specified interface have returned to normal.

Command: show interfaces tengigabitethernet 0/1/0 | include transmitted, Expected Output: transmitted 123456789 bytes, 0 underruns, 0 failures, 0 lost carrier, Description: This command verifies that the transmit statistics for the specified interface have returned to normal.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: reload original, Description: This command reinstalls the original SFP/XFP module, allowing for the restoration of service in the event of a failed repair attempt.

- Step: Reset the OSNR threshold, Command: configure terminal; interface tengigabitethernet 0/1/0; osnr threshold 20, Description: This command resets the OSNR threshold for the specified interface to its original value, ensuring that monitoring of the optical signal quality is restored.